EESC 6343 – High-Resolution Target Detection and Point Cloud Map Generation for Automotive FMCW Radar

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***Abstract*—Frequency-modulated continuous-wave (FMCW) radar is widely used in automotive sensing because it can estimate target range, relative velocity, and angle-of-arrival (AOA). This project implements and evaluates a MATLAB-based 77 GHz FMCW radar simulation inspired by the high-resolution point cloud map (PCM) generation scheme proposed by Zheng *et al.* [1]. The implemented processing chain includes FMCW waveform generation, TDM-MIMO virtual array formation, range-Doppler map (RDM) generation, multi-channel RDM fusion, improved 2D CFAR detection, post-CFAR grouping, AOA estimation, and PCM visualization. A high-resolution 9Tx/16Rx configuration synthesizing 88 useful virtual elements is compared with a reduced 4Tx/8Rx configuration producing 20 useful virtual elements. Simulation results show that the 9Tx/16Rx configuration detects all four simulated objects with accurate range, relative velocity, and azimuth estimates, while the reduced 4Tx/8Rx configuration produces additional ghost-like detections due to reduced angular resolution.**

***Index Terms*—FMCW radar, TDM-MIMO, virtual array, range-Doppler map, CFAR, angle-of-arrival estimation, point cloud map, automotive radar.**

I. INTRODUCTION

Frequency-modulated continuous-wave (FMCW) radar is widely used in automotive sensing because it can estimate target range and relative velocity simultaneously by transmitting a linearly frequency-swept chirp signal. The received echo is mixed with the transmitted signal to generate an intermediate-frequency (IF) signal, which is then processed using range and Doppler FFT operations to form a two-dimensional range-Doppler map (RDM). The RDM provides range and relative velocity information of targets in the sensing scene.

Compared with camera- and LiDAR-based sensing systems, 77 GHz FMCW radar is attractive for automotive applications because it can operate reliably under challenging environmental conditions such as fog, rain, dust, and low-light scenarios. This makes FMCW radar suitable for cruise-control, collision-avoidance, and multi-object detection tasks. However, while range and velocity estimation can be obtained from the range-Doppler processing chain, angular resolution depends strongly on the number of receive antennas and the physical aperture of the antenna array.

To improve angular resolution without physically increasing the number of receive antennas, time-division multiplexing multiple-input multiple-output (TDM-MIMO) radar can be used to synthesize a larger virtual antenna array. The reference high-resolution point cloud map generation method uses a 77 GHz FMCW radar with a 9Tx/16Rx TDM-MIMO array to synthesize 88 virtual receiver elements, achieving a theoretical angular resolution of 1.3022 degrees. The method also applies multi-channel range-Doppler fusion and an improved 2D CFAR detector to reduce masking effects and improve detection performance in multi-target environments [1].

In this project, a similar processing approach is implemented and evaluated in MATLAB. The main contributions of this coursework project are summarized as follows:

- A 77 GHz FMCW radar is simulated for a four-object automotive cruise-control scenario.
- A switchable TDM-MIMO virtual array is implemented using both 9Tx/16Rx and 4Tx/8Rx antenna configurations.
- An improved 2D CFAR detector is implemented on the fused range-Doppler map for multi-object detection.
- Point cloud maps are generated to compare the true object locations with the detected object locations.

The remainder of this paper is organized as follows. Section II describes the FMCW radar signal model and the TDM-MIMO-based virtual array model. Section III presents the process for generating the range-Doppler map. Section IV discusses the improved 2D CFAR detector and point cloud map generation. Section V describes the MATLAB simulation setup. Section VI presents the results and discussion. Finally, Section VII concludes the paper.

II. FMCW RADAR AND TDM-MIMO VIRTUAL ARRAY MODEL

A. FMCW Radar Signal Model

The transmitted FMCW chirp signal is a wideband non-stationary signal whose instantaneous frequency varies linearly with time [3]. In a typical mmWave FMCW radar, the chirp is characterized by its carrier frequency f_c, bandwidth B, chirp duration T, and frequency-modulated slope S. Texas Instruments describes this FMCW principle as a continuous frequency-modulated transmission in which the transmitted chirp is reflected by objects and then mixed with the received echo to generate an intermediate-frequency (IF) signal containing range information [2]. The transmitted FMCW chirp can be modeled as

$$x_{tx}(t) = A \cos \left(2\pi \left(f_c t + \frac{S}{2} t^2 \right) + \phi_0 \right), \quad 0 \leq t \leq T, \tag{1}$$

where A is the signal amplitude, f_c is the carrier frequency, S is the frequency-modulated slope, ϕ_0 is the initial phase, and T is the chirp duration.

The echo signal reflected from an object and received by the radar antenna can be expressed as

$$x_{rx}(t) = A_r \cos \left(2\pi \left(f_c (t - \tau) + \frac{S}{2} (t - \tau)^2 \right) + \phi_0 \right), \tag{2}$$

where A_r is the received echo amplitude and τ is the time of flight (TOF) between transmission and reception. The time delay is related to the object range by

$$\tau = \frac{2R}{c}, \tag{3}$$

where R is the object range and c is the speed of light.

After mixing the transmitted and received chirps, the output IF signal has a beat frequency equal to the difference between the instantaneous frequencies of the transmitted and received signals. For a stationary object, the beat frequency can be approximated as

$$f_b = S\tau = \frac{2SR}{c}. \tag{4}$$

For multiple objects, the received signal contains multiple delayed reflected chirps. These delayed chirps produce multiple IF tones, and a range FFT separates them into different range peaks [2]. Velocity information is obtained by transmitting multiple chirps in a frame.

B. TDM-MIMO Virtual Array

In general, the angular resolution of a radar system improves as the effective receiving aperture increases. A TDM-MIMO radar increases this effective aperture by sequentially activating multiple transmit antennas and combining each transmit-receive pair into an equivalent virtual receiver position. The angular resolution of a uniform linear array can be approximated as

$$\Delta\theta = \frac{\lambda}{N_{rx} d \cos\theta}, \tag{5}$$

where λ is the wavelength, N_{rx} is the number of receiving elements, d is the spacing between adjacent receiving elements, and θ is the target angle. The virtual array formation for both configurations is shown in Fig. 1.

C. Receiver Signal Simulation

In the TDM-MIMO scheme, the transmitters divide the channel in time and emit chirps sequentially. Between chirps, the target and radar positions are updated according to their velocities. For each sweep, the transmitted chirp propagates through a noisy free-space channel, reflects from each target, and is mixed with the transmitted signal to generate the IF signal. Following the signal model in [1], the IF signal can be written as

$$x_{IF}(n) = \sum_{i=1}^{I} A_i \cos \left(2\pi \left(S\tau_i n T_s + f_c \tau_i \right) \right) + \omega(nT_s), \tag{6}$$

where I is the number of targets, A_i and τ_i are the reflected amplitude and TOF of the ith target, T_s is the sampling period, and $\omega(nT_s)$ represents additive noise. The IF signal is then used for range and Doppler processing.

III. Range-Doppler Map Generation

A series of discrete Fourier transforms (DFTs) are used to estimate target range and relative velocity. First, the range DFT is calculated for each virtual receiver as

$$X[k_r, m] = \sum_{n=0}^{N-1} x_{IF}(n, m) \exp \left(-j \frac{2\pi}{N} k_r n \right), \tag{7}$$

where k_r is the range-bin index, m is the chirp index, and N is the number of fast-time samples. The range DFT estimates the frequency components of the IF signal, which are related to target range through the FMCW beat-frequency relationship.

Next, the Doppler DFT is applied across the chirp dimension:

$$Y[k_v, k_r] = \sum_{m=0}^{M-1} X[k_r, m] \exp \left(-j \frac{2\pi}{M} k_v m \right), \tag{8}$$

where k_v is the Doppler-bin index and M is the number of chirps. As the target moves, the phase of $X[k_r, m]$ changes across chirps. The Doppler DFT estimates this phase progression and provides relative velocity information. After all virtual receivers are processed, the result is a three-dimensional data cube,

$$Y[k_v, k_r, n_{rx}], \tag{9}$$

where n_{rx} is the virtual receiver index.

The range-Doppler maps from all virtual receivers are fused into a single 2D map as

$$Z(k_v, k_r) = \frac{1}{N_{rx}} \sum_{n_{rx}=0}^{N_{rx}-1} |Y[k_v, k_r, n_{rx}]|^2. \tag{10}$$

The fused RDM $Z(k_v, k_r)$ is then used as the input image for improved 2D CFAR detection.

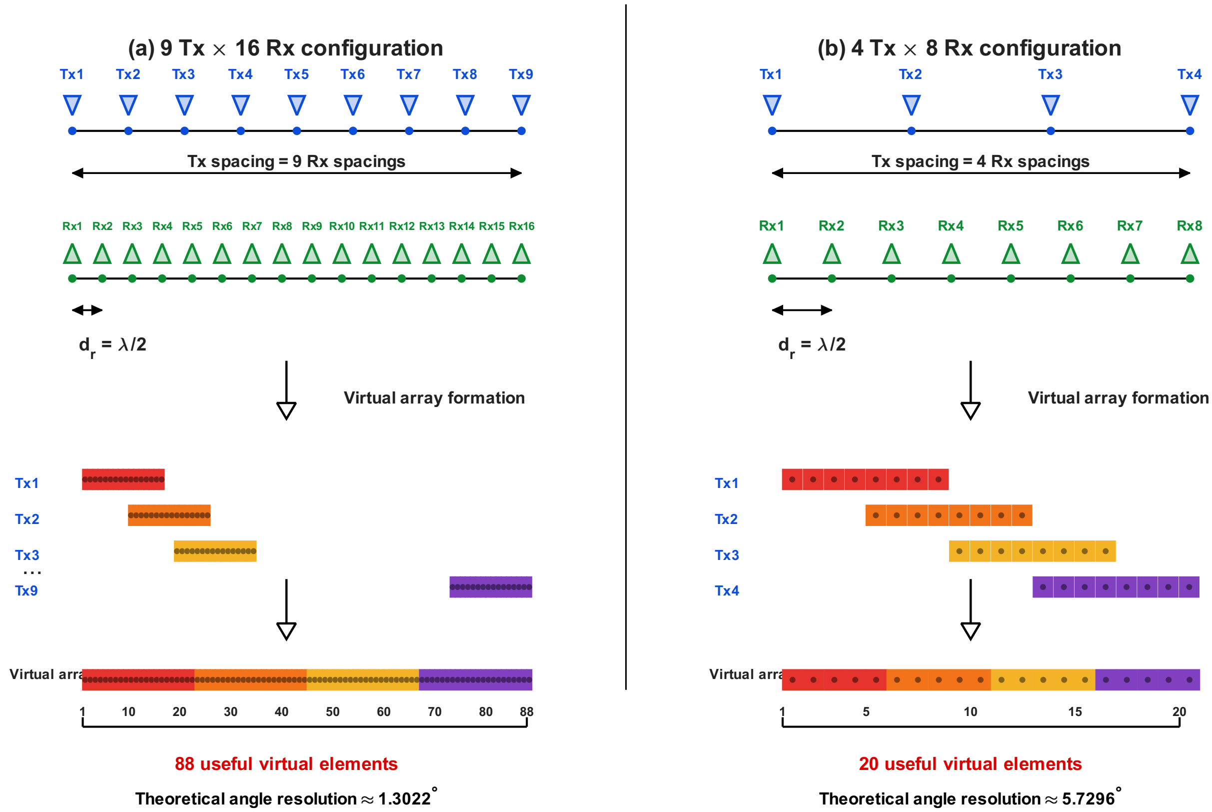

Fig. 1. TDM-MIMO virtual array formation for the 9Tx/16Rx high-resolution configuration and the reduced 4Tx/8Rx configuration.

IV. Improved 2D CFAR and Point Cloud Generation

After generating the fused range-Doppler map, an improved 2D CFAR detector is applied to identify target cells. For each cell under test (CUT), a local reference window is used to estimate the background level. The center CUT is excluded from the local average. The local noise estimate is denoted as $U(x, y)$, and the adaptive threshold is computed as

$$S(x, y) = TU(x, y), \tag{11}$$

where T is the threshold scaling factor. The detection rule is

$$Z(x, y) > S(x, y), \tag{12}$$

where $Z(x, y)$ is the fused RDM power at the CUT. If the CUT exceeds the threshold, it is marked as a detection. The detected cell is then replaced by the local noise estimate before continuing the scan. This replacement step reduces the effect of strong detected targets on the threshold estimation of neighboring cells and follows the improved CFAR logic described in [1].

After CFAR, post-CFAR grouping or non-maximum suppression is applied. The detections are sorted by RDM power, and weaker detections close to a stronger detection are suppressed. In this project, two detections are grouped if they are within 1.5 range bins and 1.5 Doppler bins. This removes duplicate detections around the same physical target peak.

For each final detection, the virtual-array response is extracted at the detected range-Doppler cell. A spatial DFT is applied across the virtual antenna elements to estimate the AOA. The final point cloud map is generated by converting range and azimuth into cross-range and range coordinates:

$$x = R\sin(\theta), \tag{13}$$

$$y = R\cos(\theta). \tag{14}$$

V. MATLAB Simulation Setup

The radar simulation was implemented in MATLAB using the Phased Array System Toolbox and Signal Processing Toolbox. The simulated scenario represents an automotive cruise-control environment, where the radar observes multiple moving objects with different ranges, velocities, azimuth angles, and radar cross sections. The simulation parameters determine the FMCW waveform parameters, including chirp duration and chirp slope. A 77 GHz FMCW waveform with a chirp duration of 6.6713 μs and a chirp slope of 22.47 MHz/μs was used as the transmitted signal. The simulated target parameters are summarized in Table I.

The radar platform speed was set to 100 km/h. Objects 1 and 2 were selected to test the system under closely spaced

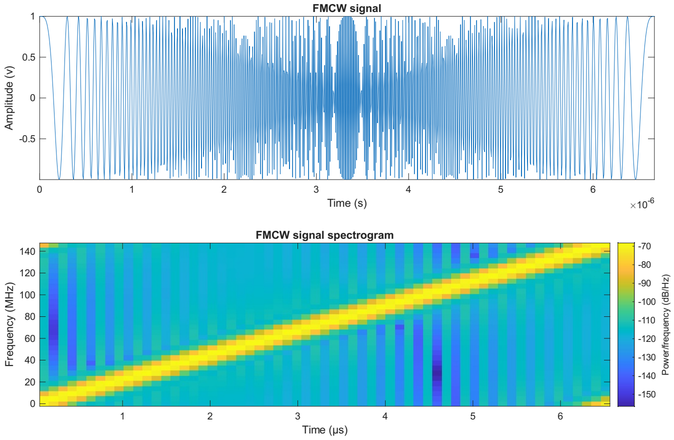

Fig. 2. FMCW signal in (a) time domain and (b) spectrogram.

TABLE I
SIMULATED TARGET PARAMETERS

Object	Range (m)	Speed (km/h)	Azimuth (deg)	RCS (m^2)
1	41	100.00	-25.000	100
2	43	100.00	-23.700	100
3	31	109.84	13.203	20
4	60	92.019	16.640	20

range and angle conditions. They have a 2 m range difference and an azimuth difference of approximately 1.3 degrees, which is close to the theoretical angular resolution of the 9Tx/16Rx configuration. Objects 3 and 4 were placed at different ranges and relative velocities to evaluate range-Doppler separation and multi-object detection performance.

Two antenna configurations were evaluated. The first configuration used 9Tx/16Rx and produced 88 useful virtual elements. The second configuration used 4Tx/8Rx and produced 20 useful virtual elements. The same target scenario was used for both configurations to compare the effect of virtual aperture size on detection and angle estimation.

VI. RESULTS AND DISCUSSION

The 9Tx/16Rx configuration produced 88 useful virtual elements and achieved a theoretical angular resolution of 1.3022 degrees. The range bin size was 0.977 m, and the Doppler bin size was 1.013 m/s. Improved 2D CFAR produced seven raw detections, and post-CFAR grouping reduced them to four final detections. This matches the four simulated objects.

The range estimates closely matched the ground-truth values. The largest range error was 0.430 m for Object 4, which is less than one range bin. The estimated azimuth angles were also close to the true angles. Objects 1 and 2 were separated in angle even though their azimuths differed by only 1.3 degrees, which is close to the theoretical angular resolution.

The relative velocity estimates were also consistent with the simulated target speeds. Objects 1 and 2 had the same speed as the radar platform and therefore had approximately zero relative velocity. Object 3 had a true relative velocity of approximately -2.733 m/s and was detected at -3.040 m/s.

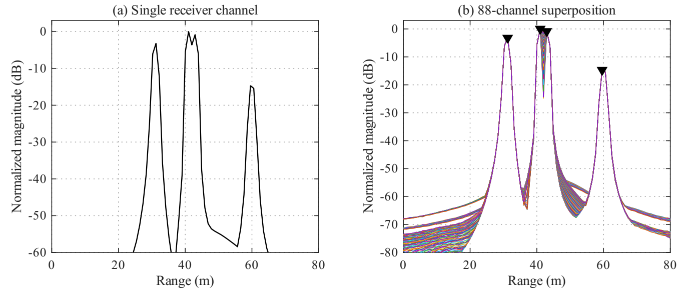

Fig. 3. Range profiles after range processing for the 9Tx/16Rx configuration: (a) single receiver channel and (b) virtual-channel superposition.

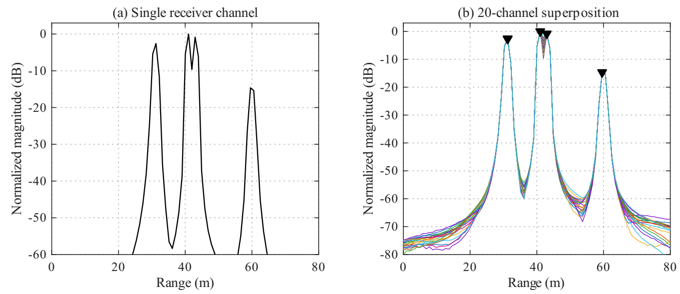

Fig. 4. Range profiles after range processing for the 4Tx/8Rx configuration: (a) single receiver channel and (b) virtual-channel superposition.

Object 4 had a true relative velocity of approximately 2.217 m/s and was detected at 2.026 m/s. These velocity errors are smaller than the Doppler bin size.

The reduced 4Tx/8Rx configuration produced 20 useful virtual elements and an angular resolution of 5.7296 degrees. In this case, improved CFAR generated 13 raw detections, and post-CFAR grouping reduced them to six final detections. Four detections corresponded to the true objects, while two extra detections appeared around the range of Object 3 but at incorrect Doppler bins. These extra detections can be interpreted as ghost-like detections caused by the reduced virtual aperture and poorer Doppler/angle separation.

These results show that the larger virtual array improves angular resolution and produces cleaner final detections. The smaller 4Tx/8Rx configuration is still able to detect the main objects, but it is more likely to produce ghost-like detections.

VII. CONCLUSION

This project implemented a MATLAB-based 77 GHz FMCW radar simulation for automotive target detection and point cloud map generation. A TDM-MIMO virtual array was used to improve angular resolution, and an improved 2D CFAR detector was applied to a fused range-Doppler map for multi-object detection. The 9Tx/16Rx configuration synthesized 88 useful virtual elements and achieved a theoretical angular resolution of 1.3022 degrees. In the simulated four-object scenario, improved CFAR produced seven raw detections, and post-CFAR grouping reduced them to four final detections, matching the number of true objects. The estimated range,

TABLE II
DETECTION RESULTS FOR 9TX/16RX CONFIGURATION

Det.	R_d (m)	R_t (m)	R_e (m)	θ_d (deg)	θ_t (deg)	v_d (m/s)
1	41.016	41.000	0.016	-25.583	-25.000	-0.000
2	42.969	43.000	-0.031	-24.148	-23.700	-0.000
3	31.250	31.000	0.250	13.137	13.203	-3.040
4	59.570	60.000	-0.430	17.185	16.640	2.026

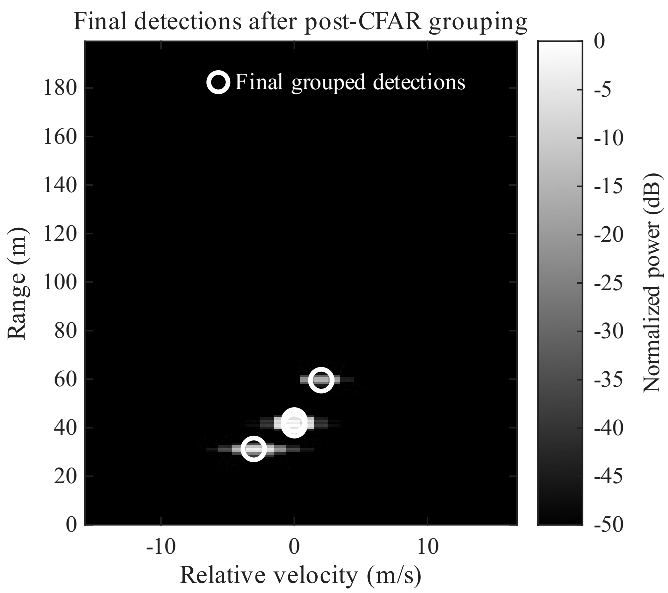

Fig. 5. Final detections after improved 2D CFAR and post-CFAR grouping for the 9Tx/16Rx configuration.

relative velocity, and azimuth values closely matched the ground truth.

The reduced 4Tx/8Rx configuration synthesized only 20 useful virtual elements and degraded the angular resolution to 5.7296 degrees. This configuration produced six final detections, including two ghost-like detections near the range of Object 3. These results show that increasing the virtual aperture improves angular resolution and reduces false or ghost-like detections. Overall, the simulation demonstrates the effectiveness of TDM-MIMO virtual array processing, fused range-Doppler mapping, improved 2D CFAR detection, and AOA estimation for high-resolution automotive FMCW radar point cloud generation.

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TABLE III
COMPARISON OF ANTENNA CONFIGURATIONS

Config.	Virtual Elements	Angle Res. (deg)	Raw CFAR Det.	Final Det.
9Tx/16Rx	88	1.3022	7	4
4Tx/8Rx	20	5.7296	13	6

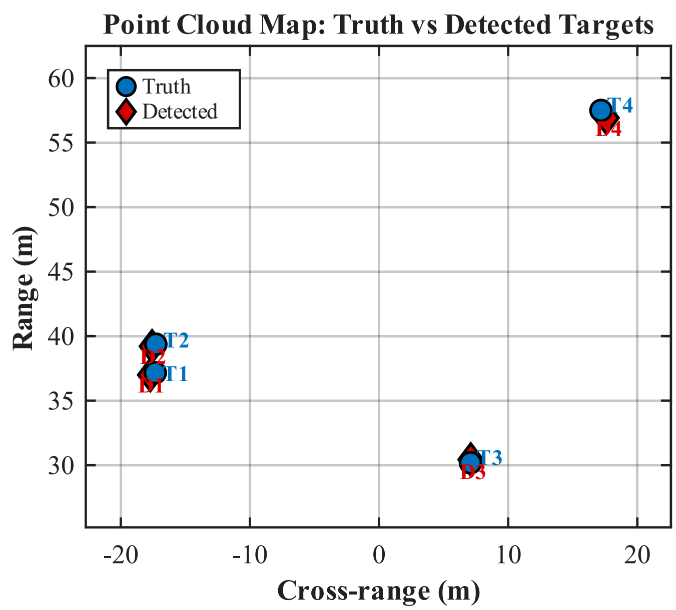

Fig. 6. Point cloud map comparing true and detected objects for the 9Tx/16Rx configuration.

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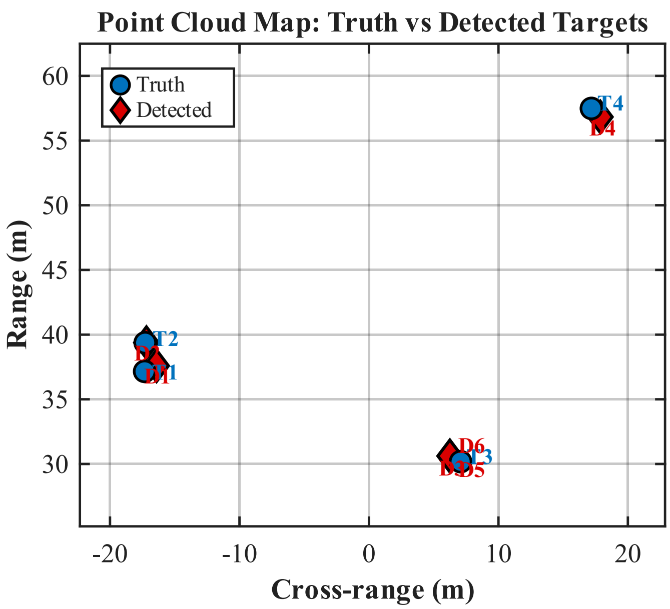

Fig. 7. Point cloud map comparing true and detected objects for the 4Tx/8Rx configuration.